BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

SECTION 'A' (Multiple Choice Questions) (50)

Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Check marks on some options in the scan are a previous solver's marks and are not transcribed. Mathematical notation is transcribed in plain text as printed.)

Note: TIME: 2 Hours — (100 Marks) — MATHEMATICS 2021 (AS PER CONDENSED SYLLABUS)

Note: NOTE: i) This section consists of 25 part questions and all are to be answered. Each question carries 02 marks. ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part. iii) The use of calculator is allowed. All notations are used in their usual meanings.

Q. SECTION 'A' (Multiple Choice Questions) (50)

Q. 1. Choose the correct answer for each from the given options:

Qi. The lim (x→0) (e^(2x) - 1)/x =

(A) 0 (B) 3

(C) 2 (D) 1

Qii. lim (x→0) sin(4/5 x)/x =

(A) 5/4 (B) 4/5

(C) 1/4 (D) 1/5

Qiii. lim (y→∞) (1 + 5/y)^y =

(A) e^5 (B) e^4

(C) e (D) 5

Qiv. The perpendicular form of the equation of a straight line is:

(A) y = mx + c (B) y - y1 = m(x - x1)

(C) x/a + y/b = 1 (D) x cos α + y sin α = p

Qv. Distance of the point (4,5) from y-axis is:

(A) 5 units
(B) 9 units
(C) 4 units
(D) 1 unit

Qvi. The distance between two points (μ cos θ, μ sin θ) and (0, 0) is:

(A) 1 units
(B) μ units
(C) μ^2 units
(D) -1 unit

Qvii. The slope of the line perpendicular to 3x - 5y - 15 = 0 is:

(A) 5/3
(B) -5/3
(C) -3/5
(D) 3/5

Qviii. The slope of the tangent to the curve $y = x^2 + 4$ at x = 1 is:

(A) 0
(B) 1
(C) 2
(D) 4

Qix. If two lines are perpendicular, then:

(A) a1a2 + b1b2 = 1
(B) a1b2 + a2b1 = 0
(C) a1a2 + b1b2 = 0
(D) a1a2 + b1b2 = -1

Qx. d/dx (x^a) =

(A) x^a ln a
(B) x^a / ln a
(C) x^a ln x
(D) ax^(a-1)

Qxi. d/dx $\sec^2 x$ =

(A) $\sec^2 x$
(B) $\sec^2 x \tan x$
(C) tan x
(D) $2\sec^2 x \tan x$

Qxii. If x = p sin t and y = q cos t, then dy/dx =

(A) -tan t
(B) q/p cos t
(C) -q/p tan t
(D) -q/p tan t

Qxiii. If n = -1, then ∫ {f(x)}^n f'(x) dx :

(A) {f(x)}^(n+1)/(n+1) + c
(B) {f(x)}^(n+1)/n + c
(C) ln f(x) + c
(D) {f(x)}^(n-1)/(n-1) + c

Qxiv. The necessary condition for f(x) to have an extreme value is:

(A) f'(x) = 1
(B) f(x) = 0
(C) f'(x) = 0
(D) f''(x) = 0

Qxv. $\int dx/(3 + x^2) =$

(A) $\tan^{-1} x/3 + c$ (B) $\sin^{-1} x/\sqrt{3} + c$

(C) $1/\sqrt{3} \tan^{-1} x/\sqrt{3} + c$ (D) $\cos^{-1} x/\sqrt{3} + c$

Qxvi. $\int e\text{^}2x\ dx$ is equal to :

(A) $2x + c$ (B) $e\text{^}(2x) + c$

(C) $e\text{^}(x^2) + c$ (D) $x^2 + c$

Qxvii. $\int (1 + x)dx/(x^2 + 2x) =$

(A) $\ln(x^2 + 2x) + c$ (B) $\ln \sqrt{(x^2 + 2x)} + c$

(C) $\ln(x^2 + 2x)^2 + c$ (D) $\ln(2x + 1) + c$

Qxviii. This equation of circle passes through the origin:

(A) $x^2 + y^2 + 8x + 7 = 0$ (B) $x^2 + y^2 - 9y + 11 = 0$

(C) $x^2 + y^2 + 8x + 11y = 0$ (D) $x^2 + y^2 - 8x + 11y + 19 = 0$

Qxix. The center of the circle $x^2 + y^2 + 6x - 10y + 33 = 0$ is:

(A) (-3, 5) (B) (-3, -5)

(C) (3, -5) (D) (3, 5)

Qxx. The radius of the circle $x^2 + y^2 + 2gx + 2fy + k = 0$ is:

(A) $\sqrt{(g^2 + f^2 + k)}$ (B) $\sqrt{(g^2 + f^2 - k)}$

(C) $\sqrt{(g^2 - f^2 + k)}$ (D) $\sqrt{(f^2 - g^2 + k)}$

Qxxi. In the parabola $y^2 = 4ax$, 4a represents:

(A) focus (B) vertex

(C) axis (D) length of latus rectum

Qxxii. The length of minor axis of the ellipse $x^2/16 + y^2/9 = 1$ is:

(A) 25 (B) 4

(C) 6 (D) 8

Qxxiii. The vertices of hyperbola $x^2/16 + y^2/4 = 1$ are :

(A) $(\pm 2, 0)$ (B) $(0, \pm 2)$

(C) $(0, \pm 4)$ (D) $(\pm 4, 0)$

Qxxiv. Magnitude of the vector $a = 3i - j + 2k$ is :

(A) 13 (B) $\sqrt{12}$

(C) $\sqrt{14}$ (D) $\sqrt{11}$

Qxxv. The unit vector in the direction of r = i + j + k is :

(A) 1

(B) √3 (i + j + k)

(C) 1/√3 (i + j + k)

(D) 1/√3 (i - j + k)

SECTION 'B' + SECTION 'C' (Subjective)

Q. SECTION 'B' (Short-Answer Questions) (30)

> Note: NOTE: Attempt any Six part questions from this section selecting at least Two part question from each questions All questions carry equal marks

Q. ANALYTIC GEOMETRY (STRAIGHT LINE) AND VECTOR ALGEBRA

Q. ANALYTIC GEOMETRY (CONIC SECTIONS)

Q. CALCULUS

Q. SECTION "C" (Detailed Answer Questions) (20)

> Note: NOTE: Attempt Two questions from this section.

> Note: نوٹ: سال 2020 میں گیارہویں اور بارہویں جماعت کے امتحانات نہیں ہوئے تھے۔

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